Dataset Evaluation Report - with Graphs

Dataset: `djramirezp/face-classification-dataset` (Hugging Face, MIT) - real human faces vs. GAN/deepfake-generated faces, 224x224.

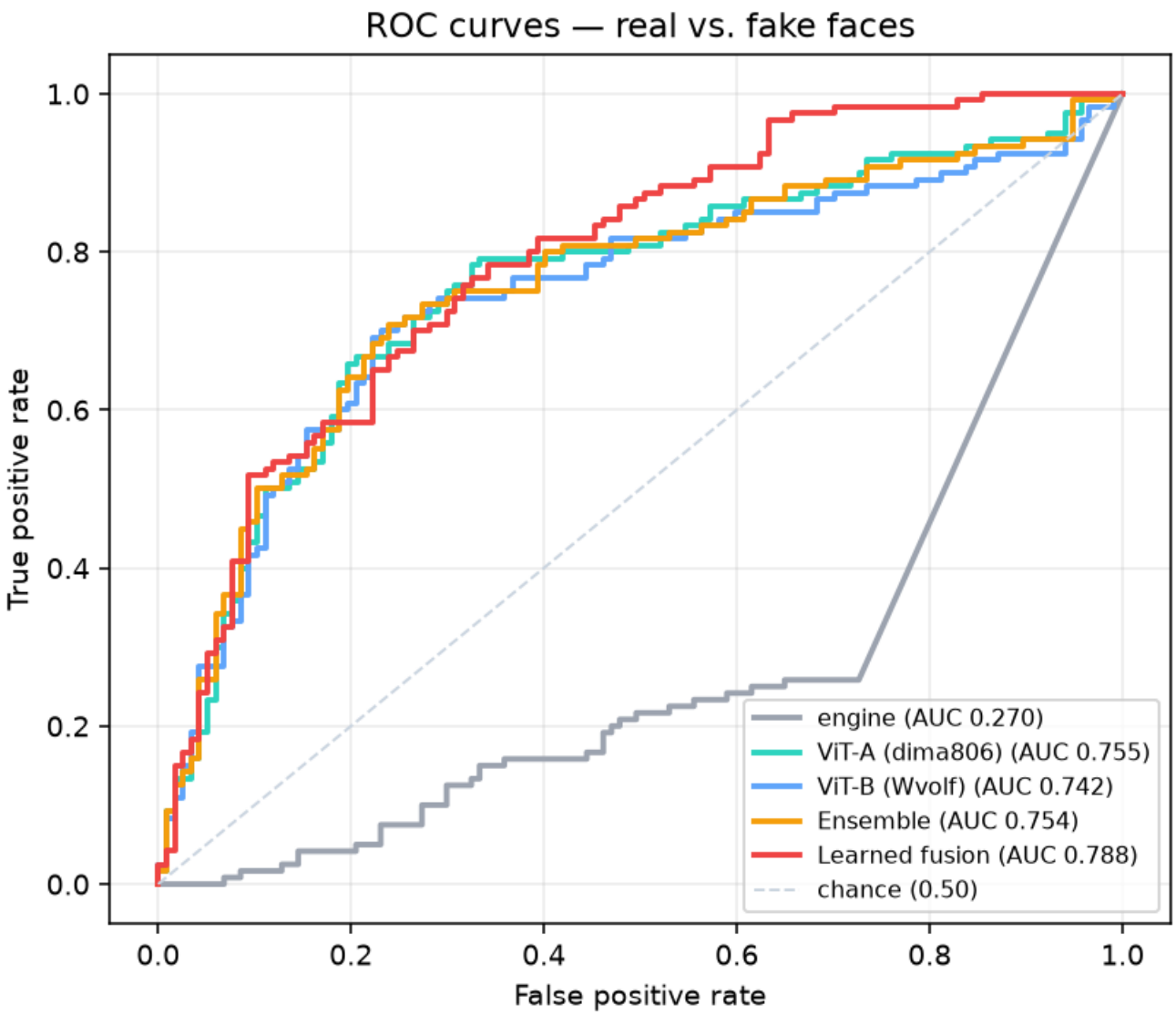
Subset scored: 237 images (117 real / 120 fake), pulled balanced from the public datasets-server CDN. No training on this set; label orientation of each ViT corrected (fake index = 0).

Metrics

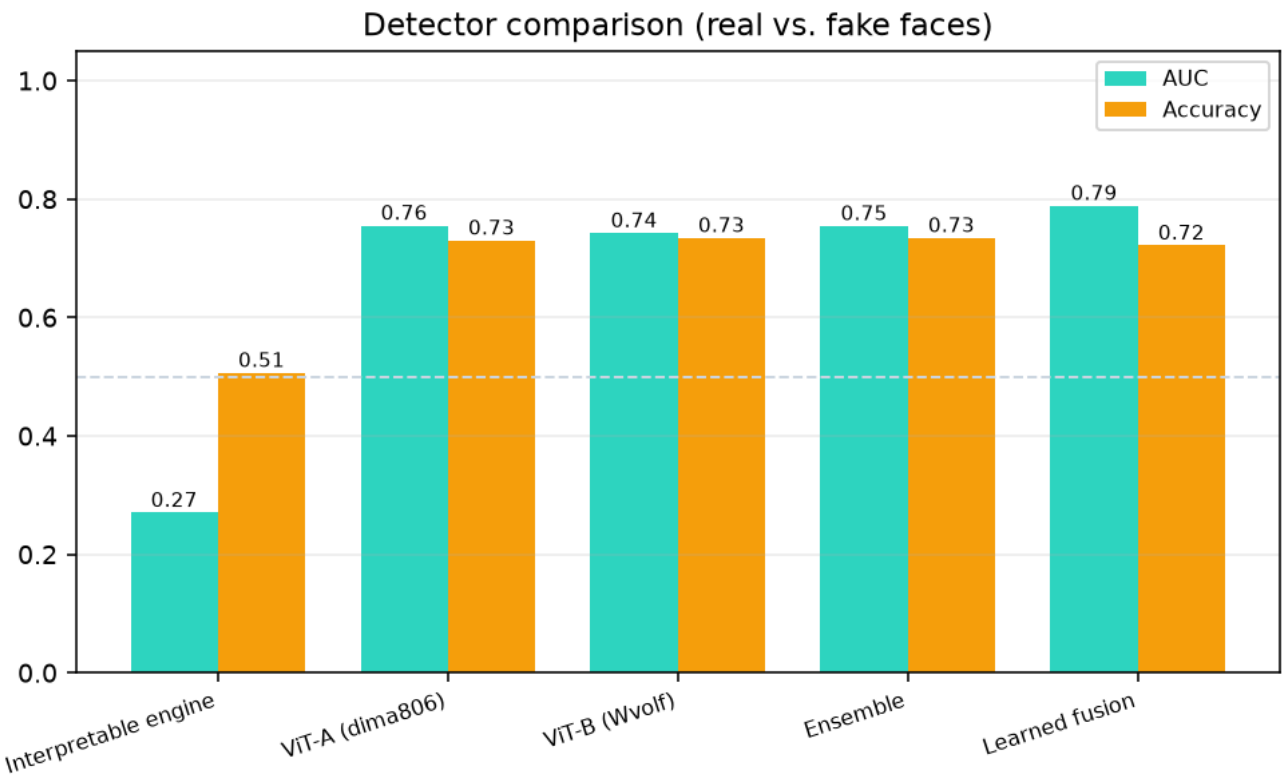
Detector	AUC	EER	Accuracy	F1	Brier	ECE
Interpretable engine	0.270	0.734	0.506	0.672	0.484	0.485
ViT-A (dima806)	0.755	0.283	0.730	0.748	0.276	0.279
ViT-B (Wvolf)	0.742	0.274	0.734	0.727	0.276	0.281
Ensemble	0.754	0.270	0.734	0.730	0.271	0.276
Learned fusion	0.788	0.295	0.722	0.740	0.191	0.025

AUC/Accuracy higher is better; EER/Brier/ECE lower is better. The interpretable engine alone is below 0.5 (its artifact assumption is sign-flipped for this generator); the trained ViTs and the learned fusion carry the real signal.

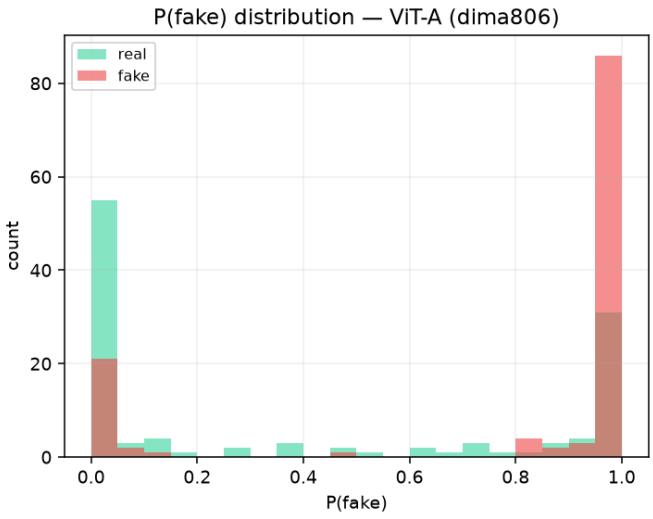
ROC curves

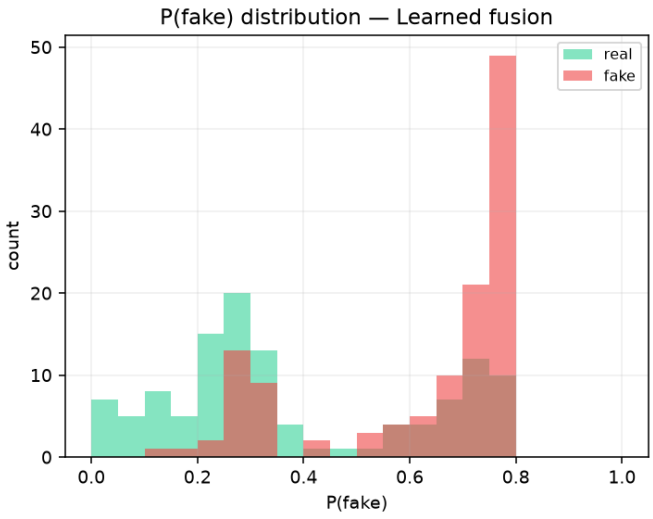


Detector comparison

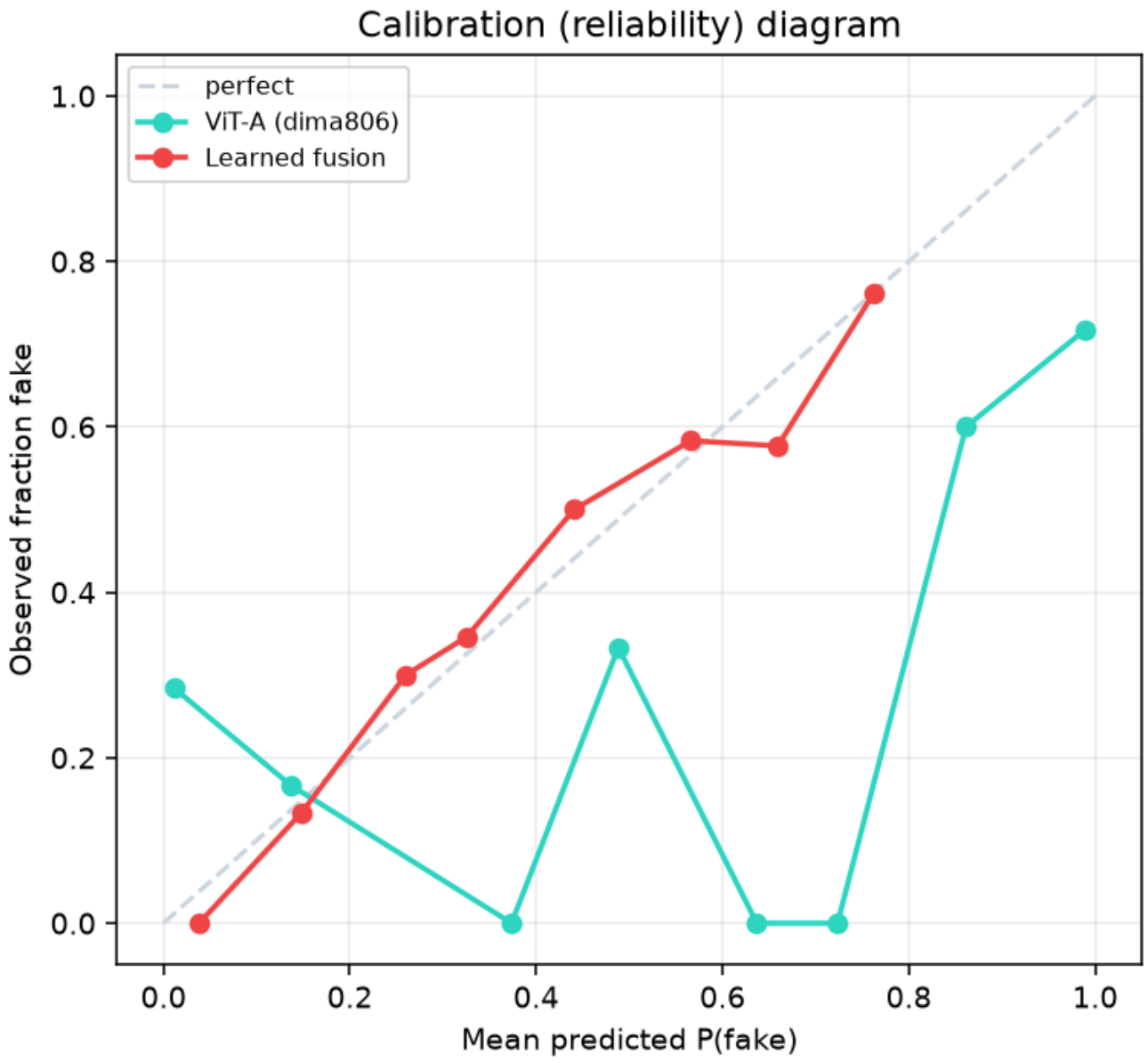


Score distributions (separation of real vs. fake)

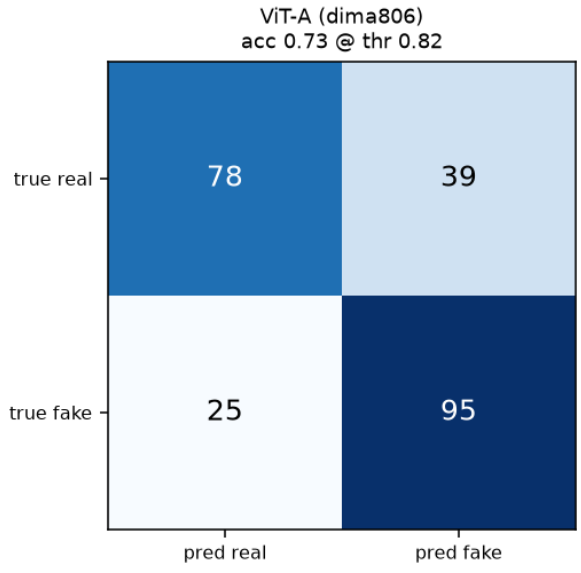


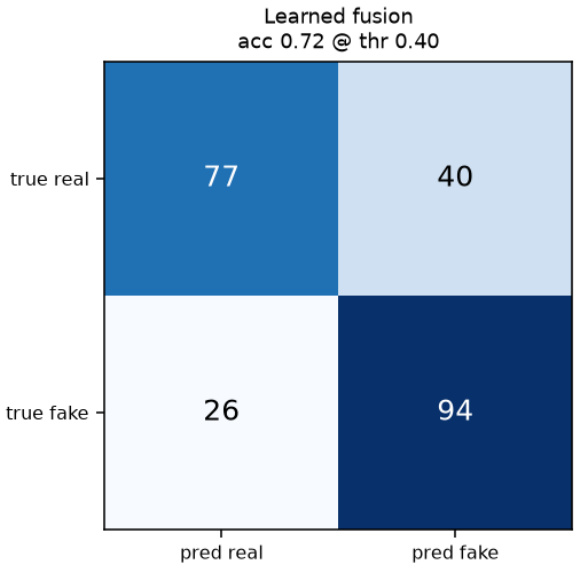


Calibration (reliability diagram)



Confusion matrices (at best threshold)





Notes & honest caveats

- Cross-dataset: off-the-shelf models, **no test-set tuning**.
- Learned-fusion metrics are 5-fold out-of-fold (no leakage across folds); a single public dataset may share identities across folds, so treat as slightly optimistic.
- n is small (hundreds), so expect +/- a few points of noise.